PAPER

# Locating and Mitigating Factual Hallucination Circuits in Small Language Models via Activation Patching

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**SUMMARY** Small language models (SLMs) are increasingly deployed in resource-constrained systems, yet they remain prone to factual hallucination, particularly when a misleading context conflicts with parametric knowledge (a context–memory conflict). We present a systematic activation-patching study of this conflict in three SLMs: Gemma-2-2B (2,000 facts), Llama-3.2-1B (600 facts), and Qwen2.5-1.5B (600 facts), each filtered to facts the model demonstrably knows. Restoration effects concentrate in the final quarter of the network across all three architectures (pairwise Spearman rank correlation of layer profiles 0.81–1.00), with a small set of late-layer attention heads dominating. Bidirectional patching shows that influential heads act in both patching directions (head-level superposition) and separates dual-role workhorses from conflict-specialist heads whose neutral-control effect is minimal. In Gemma-2-2B, knocking out the single specialist head L22.H4 reduces misleading-context distractor-following by 35% relative (exact McNemar $p=3.4\times10^{-23}$) while increasing factual accuracy by 5.3 percentage points, with MMLU retention 0.998, a 5% perplexity cost, and accuracy under a supportive correct context nearly intact (96.1% vs. 98.2%); single-head knockouts from the patching ranking reduce distractor-following by 30–50% in the other two models as well. In contrast, norm-based selection *increases* distractor-following; random single heads also reduce it, and the one measured for capability collapses MMLU (retention 0.69 vs. 0.998). We report quantified dose–response trade-offs between reduction and capability, with relation-level splits guarding against circular analysis.

***key words:*** *mechanistic interpretability, hallucination, knowledge conflict, activation patching, attention heads*

## 1. Introduction

Language models of 1–2.6B parameters now power on-device assistants, retrieval-augmented pipelines [9], and document-processing systems where latency and memory budgets preclude larger models. In such deployments the model must frequently integrate an external context with its parametric knowledge. When the context asserts a false fact—whether through a retrieval error, an adversarial document, or a mistaken user; in effect, in-context misinformation [7]—the model faces a *context–memory (CM) conflict* [5], and resolving it incorrectly produces a factual hallucination, which can then cascade into further errors [32].

Why a mechanistic account? Deployed systems need interventions that are cheap, targeted, and auditable: knowing *which* components carry the conflict lets a builder trade hallucination against capability with measured confidence. The account is more valuable still if the responsible components transfer across architectures, which is why replication is a first-class concern here.

A growing body of work studies hallucination mechanisms, including knowledge-deficit failures [2] and probes/directions for truthfulness [3], [4]. Head-level analyses of CM conflict exist in text-only models [17], [18] and multimodal settings [19]. What has been missing for this size class is a systematic causal account—which components, located how, trade hallucination against capability when intervened upon, and whether findings replicate across architectures—and, for head-level work generally, a neutral-control separation of head roles and quantified single-head intervention costs.

This paper provides that account with a design that separates localization from validation: heads are selected only on a development split of relations, and all intervention results are reported on held-out relations. Our contributions are:

- **Localization.** A coarse-to-fine activation-patching map of the CM-conflict circuit in Gemma-2-2B (and its replication in two other architectures): restoration effects concentrate in late layers, and the attention/MLP decomposition shows the conflict is attention-dominated, unlike the MLP-heavy picture of classical fact recall.
- **Characterization.** Bidirectional patching quantifies superposition per head and separates dual-role heads from conflict-specialist heads with minimal neutral-control effect.
- **Quantified intervention trade-offs.** Dose–response curves for knockout and mean-ablation, with fairness baselines: patching-guided selection reduces hallucination significantly (McNemar $p < 10^{-6}$ even for a single head), whereas activation-norm selection *increases* it; capability costs are dissociated (MMLU retention $\approx 1.00$; perplexity +5–20% across interventions).

## 2. Related Work

**Hallucination and knowledge conflict.** Surveys taxonomize hallucination in NLG [6] and knowledge conflicts in LLMs [5]. Behavioral studies quantify when context overrides memory [8]; Chuang et al. detect contextual hallucinations from attention maps [10], and attention-node analysis likewise enables quality-improving masking in machine translation [11]. Data-side remedies refine noisy factual resources with LLM assistance [12]. Yu et al. trace non-factual

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hallucination to knowledge deficits in MLPs and selection failures in attention [2]; our corruption regime instead targets *knows-but-overridden* failures. Mechanistic work has also examined conflicts inside parametric memory itself [23] and cautioned that heads appearing to implement factual recall may partly implement generic copying [24]—a confound our neutral-control design quantifies directly.

**Activation patching and circuits.** Causal mediation introduced the method [13]; ROME localized factual associations with causal tracing [1]; best practices are systematized in [14]. Head-level analyses include IOI [15] and copy suppression [16]. Closest to us are two head-level studies of knowledge conflict. JuICE reports that influential heads play superposed dual roles in knowledge conflicts [17]; PH3 identifies memory and context heads by path patching and prunes them to steer conflict resolution in either direction, demonstrating cross-model, cross-relation, and cross-format transfer [18]. Our work differs in what is selected and how the cost is measured: (i) a neutral-control analysis that separates conflict specialists from dual-role heads, a distinction neither method draws; (ii) single-head specialist knockout selected and evaluated under a relation-disjoint dev/held-out split; and (iii) explicit capability trade-off curves (MMLU, perplexity) reported alongside hallucination metrics, which, to our knowledge, neither study reports. Our top-$k$ knockout is structurally close to PH3-style context-head pruning; specialist selection removes 35% of distractor-following with one head while accuracy rises, whereas aggregate pruning trades accuracy for compliance (Sect. 4.3).

**Inference-time intervention.** ITI steers along probe-identified truthful directions [3]; truth directions are linearly represented [4]; hallucination can also be detected from sampling uncertainty [22]. Our interventions are *subtractive* (head knockout/mean-ablation) and selected by causal patching rather than probes, with explicit capability trade-off curves.

**Attention sinks and knowledge editing.** Autoregressive models allocate disproportionate attention to initial tokens ("attention sinks"), which stabilize generation but are usually treated as functionally inert no-ops [20]. Our findings add a twist: the conflict-specialist heads that dominate the CM-conflict circuit are BOS-sink heads that are causally *active* under misleading context (Sect. 4.2). Separately, knowledge editing locates and rewrites stored facts [1], [21]; we instead locate components involved in *context override* and ablate them at inference time, leaving parameters untouched.

## 3. Experimental Setup

Fig. 1 summarizes the protocol.

**Models.** Primary: Gemma-2-2B (26 layers, 8 heads) [27]. Cross-model: Llama-3.2-1B (16L/32H) [28], [29], Qwen2.5-1.5B (28L/12H) [30]. All experiments use bfloat16 on a single 24GB GPU; a parity check between our hooking framework and reference HuggingFace logits passed in float32 (mean KL $1.3 \times 10^{-5}$, argmax agreement 100%). Precision robustness of the metrics themselves is reported in Sect. 4.1.

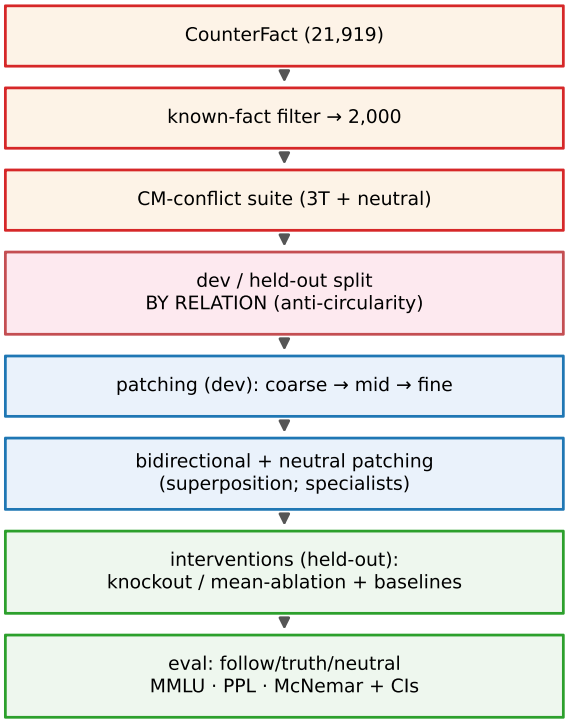



**Fig. 1** Protocol overview. Data construction (top), causal localization on the dev split (middle), and interventions evaluated exclusively on held-out relations (bottom).

**Data.** From CounterFact [1] we retain facts each model demonstrably knows (first-token argmax over the full vocabulary equals the first token of the space-prefixed true target, and the first-token logit gap over the counterfactual distractor $\geq 0.5$), sampling 2,000 facts (16.7% survival of 12,000 scored) split 807/1,193 dev/held-out *by relation*, so the held-out split shares no relation with the split used for any selection. A CM-conflict suite asserts the distractor in three templates; a neutral control replaces it with a non-answer token.

**Metrics and definitions.** Let $z_t(x)$ and $z_d(x)$ denote the logits, at the answer position of input $x$, of the first tokens of the true target and the distractor, respectively. The *logit difference* is $\mathrm{LD}(x) = z_t(x) - z_d(x)$. Let $c$, $m$, and $n$ denote the clean, misleading, and neutral prompts for a given fact, and let $h$ index a specific attention head. Writing $x \leftarrow y_h$ for the input $x$ with head $h$'s activation replaced by its value from input $y$, the three directed patching effects are:

$$D = \mathrm{LD}(c) - \mathrm{LD}(m), \tag{1}$$

$$A_h = \frac{\mathrm{LD}(m \leftarrow c_h) - \mathrm{LD}(m)}{D}, \tag{2}$$

$$B_h = \frac{\mathrm{LD}(c) - \mathrm{LD}(c \leftarrow m_h)}{D}, \tag{3}$$

$$N_h = \frac{\mathrm{LD}(n \leftarrow c_h) - \mathrm{LD}(n)}{\mathrm{LD}(c) - \mathrm{LD}(n)}. \quad (4)$$

Here $A_h$ is the *restore effect* (how much patching the clean value into the misleading run restores the true-answer margin; $A = 1$ is full restoration), $B_h$ is the *context-push effect* (how much patching the misleading value into the clean run degrades the margin; $B = 1$ is full degradation), and $N_h$ is the *neutral effect* (patch into the neutral-context run, normalized by the clean–neutral gap, *not* the clean–misleading gap $D$ of Eq. (1) used for $A$ and $B$). The *directional asymmetry index* $S = |A - B|/(|A| + |B| + \epsilon)$, with $\epsilon = 10^{-8}$, equals 1 for a head that pushes one direction only and 0 for a balanced dual-role head; low $S$ thus corresponds to strong superposition in the sense of [17], and we use the two framings interchangeably below. When both $|A|$ and $|B|$ are near zero the index is uninformative, and we exclude such heads from superposition-based claims. Effects where the denominator has magnitude $< 10^{-6}$ are set to 0. Head-level values of $A$, $B$, and $N$ are means over per-fact effects; the reported neutral ratio $N/A$ is the ratio of these means (not a mean of per-fact ratios). Hallucination rate is the fraction of misleading-prompt items whose greedy completion equals the distractor (context-following); accuracy is the fraction matching the true target—both are first-token measures, so "accuracy" means first-token accuracy throughout. We use the term "misleading-context distractor-following" for this specific, programmatically verifiable metric, which is a strict subset of open-ended factual hallucination. We use "circuit" to mean the causally localized head set, not a fully traced pathway.

**Protocol.** Stage 1 (coarse): patch residual-stream block outputs layer-by-layer under two corruptions—ROME-style embedding noise, and the misleading context itself—measuring the normalized restoration of the true-vs-distractor logit difference. Stage 2 decomposes attention vs. MLP outputs on top layers; Stage 3 patches per-head $z$-vectors. Stage 4: bidirectional and neutral-context patching characterize each top head ($A$, $B$, $N$ in Eqs. (2)–(4); $S$ as above). Stage 5 intervenes by zeroing or mean-ablating selected heads and evaluates hallucination rate, accuracy, and neutral accuracy on held-out items; MMLU (2,000-question fixed subset) and WikiText-2 perplexity (1024-token windows) measure capability under the same hooks. Fairness baselines ablate randomly selected heads (5 seeds) and top-$|z|$-norm heads. Statistics: fact-clustered bootstrap 95% CIs (10,000 resamples); two-sided exact binomial McNemar tests against "none", where $b$ counts items the intervention fixes and $c$ items it breaks (per outcome: distractor-following, truth, neutral, or correct-context accuracy). The prespecified primary comparison is L22.H4 vs. none; all other configurations are exploratory.

**Reproducibility details.** Misleading templates (distractor $d$, prompt $p$): T0 = "*Earlier you learned that the correct answer is d. p*"; T1 = "*The document states: p d. Now answer correctly: p*"; T2 = "*Hint from the user: I am sure it is d. p*". Neutral control replaces $d$ with "unknown". The relation-level split uses seed 0: 31 relations, 16 dev (807 facts) and 15 held-out (1,193), zero overlap. Answers are scored by first-token argmax at the final position against the first token of the space-prefixed targets (no autoregressive generation, hence no whitespace/punctuation/alias handling). All retained targets are single words and no pair shares a first word, so the comparison is well-defined; known-fact status uses the same level. A greedy 8-token completion check on 300 held-out items reproduces first-token accuracy exactly (none 82.3%, L22.H4 83.7%), so first-token and full-answer scoring coincide on this dataset. Patching positions: the noise regime patches every non-BOS position of the identically-lengthened runs; the misleading regime and head-level patching use the last three tokens, right-aligned across runs. Noise corruption is Gaussian at the embedding layer ($\sigma = 3\times$ embedding std, all non-BOS tokens, one draw per case). The top-norm baseline ranks heads by mean $|z|$ over 64 dev bare-fact prompts; the mean-ablation vector is the mean $z$ over the same prompts. Six of the top-10 restore heads pass $|N|$<0.01; L22.H4 is the highest-$B$ qualifier, spec3 the three highest-$A$ qualifiers (L22.H4, L22.H0, L20.H3)—rank-based choices, no further threshold. Cross-model profile correlation interpolates each model's per-layer mean-effect profile at 10 evenly spaced depth fractions (0.05–0.95) before Spearman ranking. Models use HuggingFace checkpoints `google/gemma-2-2b`, `meta-llama/Llama-3.2-1B`, `Qwen/Qwen2.5-1.5B` (base, not instruct-tuned) loaded in bfloat16 via TransformerLens 2.18 with `center_unembed=False`, `fold_ln=False`. Head ablation zeroes the head's $z$-vector at *all* token positions. MMLU uses the `cais/mmlu` "all" test split, a fixed 2,000-question subset (seed 0), 0-shot, scored by comparing logits of the letters "A"–"D" at the answer position [25]; WikiText-2 uses the `wikitext-2-raw-v1` test split, 100 windows of 1024 tokens [26]. The parity check between TransformerLens and HuggingFace logits was conducted in float32 (mean KL $1.3\times10^{-5}$); all experiments run in bfloat16. KL quantifies distribution-level divergence between precisions, not the error of our summary metrics directly, so we additionally compared logit differences, patching effects, and head rankings between the two precisions on a fixed dev sample (Sect. 4.1).

## 4. Results

### 4.1 Behavioral landscape and localization

On known facts, bare-prompt accuracy of Gemma-2-2B is 100%. Under misleading context the competition is nearly balanced (47.3% following, 95% CI 46.0–48.5, vs. 46.2% truth) but strongly template-dependent (24.2%/98.4%/19.2%), matching reported persuasion-strength variation [8]; the neutral control stays at 87.4% truth with 1.2% neutral-token copying (Table 1). Template 1 (distractor adjacent to the repeated question) drives following to 98.4%—an upper bound on recency/copy contributions consistent with, though not proof of, a copy path-

**Table 1** Behavioral landscape on known facts (Gemma-2-2B, $n$=2,000; misleading suite per template).

| condition | context-following | truth |
|---|---|---|
| clean (no context) | — | 100% |
| misleading T0 | 24.2% | 66.4% |
| misleading T1 (adjacent distractor) | 98.4% | 1.1% |
| misleading T2 | 19.2% | 71.0% |
| neutral control | 1.2%† | 87.4% |

†fraction copying the neutral token.

**Table 2** Block-level decomposition (mean restoration effect, top layers).

| block | noise regime | misleading regime |
|---|---|---|
| L22 attention | 0.074 | **0.121** |
| L25 attention | 0.018 | 0.048 |
| L21 MLP | **0.076** | ≤0.01 |
| L20 MLP | 0.058 | ≤0.01 |

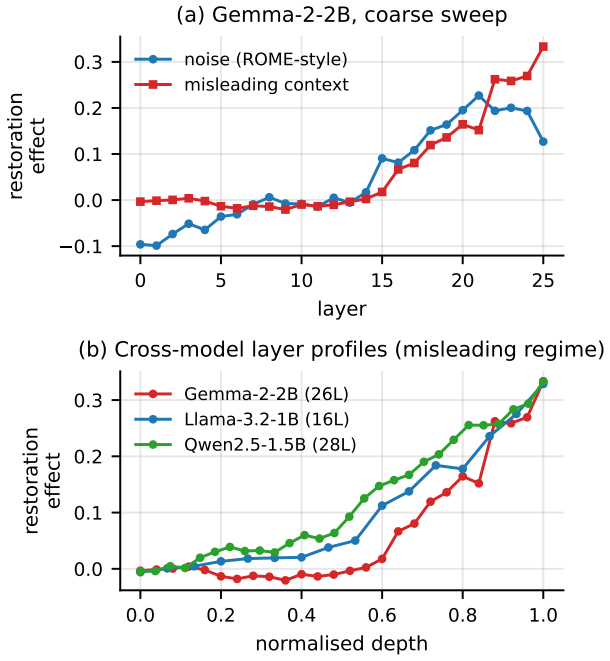



**Fig. 2** Layer profiles of restoration effect. (a) Gemma-2-2B under both corruption regimes; (b) misleading-regime profiles across the three architectures at normalized depth. Late-layer concentration replicates (pairwise Spearman 0.81–1.00).

way. Reference benchmarks (lm-eval-harness, full MMLU test set): MMLU 49.4%±0.4, TruthfulQA MC1/MC2 [31] 24.5%/36.3%, WikiText-2 PPL 65.3 (1024-token windows). The 2,000-question MMLU subset used for intervention evaluation yields 50.0% at baseline; the two scores differ because of subset sampling, not protocol error. **Precision robustness.** Repeating the LD measurements and top-20 head-effect ranking in float32 vs. bfloat16 on the same 200 dev facts: mean |ΔLD| is 0.07–0.08 logits vs. mean |LD| 2.7–8.7 (0.9–3%); head effects rank-correlate at Spearman 0.99 (max per-head difference 0.008), and L22.H4 ranks first among the compared heads under both precisions.

Fig. 2 shows the coarse maps. Restoration under the misleading regime concentrates in the final layers, peaking at the last layer (0.333); the noise regime peaks earlier (layer 21, 0.227). Table 2 decomposes the top blocks: the misleading-conflict signal is attention-dominated (L22 attention 0.121 vs. ≤0.01 for every MLP block), whereas pure recall restoration under noise mixes MLP and attention almost evenly. (The regimes also differ in patched positions—all non-BOS for noise, last three for the conflict—so this contrast is specific to the protocol; a matched-position control is future work.) This contrast is itself informative: the classical ROME-style picture of factual recall as MLP-stored knowledge [1] does not transfer to the conflict regime, where what an intervention can restore flows through attention. Head-level patching then identifies a compact set: {L22.H0 (0.26), L25.H4 (0.24), L20.H3 (0.15), L25.H6 (0.12)} under noise, and {L25.H4 (0.08), L22.H4 (0.07), L22.H0, L20.H3} under the misleading context—note the partial overlap: three of four noise-top heads reappear in the conflict ranking, and L22.H4 emerges only under conflict.

### 4.2 Superposition and conflict specialists

Bidirectional patching (Fig. 4) shows every influential head conveys the misleading context more strongly than it restores the memory ($B > A$; asymmetry indices $S$ 0.21–0.62), consistent with head-level superposition [17] in this regime. The neutral control separates two populations, subject to a normalization caveat: $N$ divides by the clean–neutral gap, which is far smaller than $D$ (median 0.25 vs. 6.7 logits; 35% of cases below 0.5 logits, and 5.3% with $|\text{gap}|<10^{-6}$ set to $N$=0), so the ratio $N/A$ must not be read as a fraction of effect retained—in raw logit units, patching L25.H4's clean value changes the neutral run by only 0.01 logits (0.50 logits when patched into the misleading run), and no top head shows a large absolute neutral effect. Read instead as each head's *share of the clean–neutral gap*, L25.H4 carries the largest share among top heads and keeps it under every denominator threshold we tested (0.5, 1, 2 logits), whereas L22.H4, L22.H0, and L20.H3 remain at $N \approx 0$ under every threshold—*conflict specialists* whose minimal neutral-control effect is robust to the normalization choice. Fig. 3 measures, from the answer position of 32 misleading held-out prompts (mean±SD), where each head reads. The strongest specialist, L22.H4, concentrates 68 ± 9% of its answer-position attention on ⟨bos⟩ and a further 18 ± 8% on the token asserting the distractor (86% combined); the other specialists split 22–35% BOS / 10–12% distractor, whereas the dual-role heads L25.H4/L25.H6 read the distractor token only 3%/0.5%. Reading-position attention to the distractor is modest for every head (2–11% averaged over positions following the distractor). We note that BOS precedes the misleading context in the causal decoder, so BOS attention limits *what the head reads* rather than explaining *how* it conveys context information—the mechanism may operate

**Table 3** Per-head characterization (Gemma-2-2B, top-10 by restore effect $A$; $S$ = directional asymmetry index; $N/A$ = neutral ratio).

| head | $A$ | $B$ | $N$ | $S$ | $N/A$ |
|---|---|---|---|---|---|
| L25.H4 | 0.081 | 0.134 | 0.062 | 0.24 | 0.76 |
| L22.H4 | 0.071 | 0.143 | −0.006 | 0.34 | −0.09 |
| L22.H0 | 0.048 | 0.095 | 0.002 | 0.33 | 0.05 |
| L20.H3 | 0.038 | 0.096 | −0.004 | 0.43 | −0.10 |
| L23.H7 | 0.016 | 0.038 | −0.012 | 0.41 | −0.72 |
| L23.H4 | 0.012 | 0.022 | −0.013 | 0.29 | −1.02 |
| L20.H7 | 0.010 | 0.042 | −0.009 | 0.62 | −0.91 |
| L22.H3 | 0.010 | 0.030 | −0.013 | 0.50 | −1.33 |
| L25.H2 | 0.010 | 0.024 | 0.004 | 0.42 | 0.38 |
| L21.H1 | 0.009 | 0.019 | 0.001 | 0.34 | 0.06 |

through the value vectors at BOS, through attention reallocation, or through downstream residual-stream effects; separating these hypotheses remains open. Table 3 reports the full per-head characterization for the ten highest-restore heads. Two structure-level observations follow. First, $B > A$ for *every* head—no influential head is a pure "memory carrier"; all participate in conveying the context. Second, the specialist cluster (L22.H4, L22.H0, L20.H3) combines $N \approx 0$ with mid-range asymmetry ($S$ 0.33–0.43), whereas heads combining large $|N/A|$ with higher asymmetry (L20.H7, L22.H3) sit lower in the restore ranking: a large generic role, not asymmetry alone, makes a head a poor intervention target.

### 4.3 Interventions and quantified trade-offs

Table 4 and Fig. 5 report the held-out interventions. Knocking out the dev-ranked top-$k$ heads reduces context-following monotonically (19.3%→1.6% at $k$=20) at matching accuracy cost. The Stage-4 taxonomy, however, predicts that removing a conflict specialist should suppress the misleading pull at limited collateral cost. This is borne out: knocking out the single head L22.H4 reduces hallucination to 12.6% (two-sided exact McNemar $b$=81, $c$=1, $p$=3.4×10$^{-23}$) while *increasing* accuracy to 79.7% (+5.3 points; the gain is itself significant: $b$=81, $c$=18, $p$=1.0×10$^{-10}$) and leaving neutral accuracy at 92.0% (baseline 93.5%; the 1.5-point decrease is small but significant, $b$=5, $c$=23, $p$=9.1 × 10$^{-4}$). Fact-clustered 95% CIs for L22.H4: follow 10.7–14.5%, truth 77.5–82.0%, neutral 90.4–93.5%. By contrast, ablating the five highest-norm heads *increases* hallucination to 33.0% ($b$=9, $c$=173): these are early/mid-layer high-$|z|$ heads (L2–L16), not the late-layer heads of Table 3, and their removal tips the final balance toward the context. Random-head ablation is erratic across seeds (follow 12.6–22.2%, truth 44.8–78.8% over five seeds, with inconsistent sign and magnitude).

**How special is the specialist?** A uniformly random *single*-head knockout reduces following for every seed tested (5.2–12.2%; truth 81.9–88.3%), and layer-matched late-layer random heads behave the same (9.3–10.3%): reducing distractor-following by one-head $z$-knockout is *not* unique to the specialist—the conflict signal is distributed, so removing almost any single head weakens it. The specialist is distinguished by cost in the one random head we measured for capability: the behaviorally best random seed (L17.H3: follow 5.2%, truth 88.3%) collapses MMLU to 34.6% (retention 0.69 vs. 0.998 for L22.H4). Capability was not measured for the other random heads, so this case—together with random-5's erratic truth rates—illustrates, rather than bounds, the cost profile of unprincipled selection.

Capability costs are dissociated (Table 5): MMLU retention remains 0.99–1.01 for every tested intervention, and a paired per-question rerun for L22.H4 detects no change on this subset (paired difference −0.40 points, 95% CI [−1.10, +0.30]; $b$=22 gained, $c$=30 lost; exact $p$=0.33)—no detected change, not demonstrated equivalence—whereas perplexity rises 5–20%. The single-head specialist intervention is the cheapest of all: a 35% relative hallucination reduction—with an accuracy *gain*—at MMLU retention 0.998 and a perplexity cost of only 5%.

**Correct-context control.** A correct-context suite (T0 asserting the *true* target, same held-out facts) tests damage under context that is actually right. Baseline is near ceiling (98.2% truth); L22.H4 leaves 96.1% (small but significant, $b$=0, $c$=25, $p$=6 × 10$^{-8}$; far below its 5.3-point misleading-context gain), whereas spec3 (81.6%) and top-5 (60.2%) destroy it. Because this context concerns already-known facts, the result bounds damage under consistent context rather than measuring context *use* per se; deployment beyond this counterfactual regime (e.g., general RAG reliability) is out of scope.

**Template robustness.** Re-evaluating the none, L22.H4, and top-5 conditions on the two remaining held-out templates tests whether the specialist's effect depends on how the context asserts the distractor. On T2 (user-hint template) L22.H4 again reduces following (16.6%→11.1%) while raising accuracy by 3.3 points, replicating T0. On T1—the extreme adjacent-distractor template whose baseline behavior is nearly pure context-following (98.4%) and which the neutral-control analysis attributes largely to recency copying—L22.H4 reduces following from 98.4% to 89.4% (9.0 percentage points; 9% relative) and raises accuracy from 0.8% to 8.7%, while top-5 reduces following from 98.4% to 58.3% (40.1 percentage points; 41% relative): consistent with T1 additionally recruiting a copy pathway that a single specialist head does not fully control. The intervention thus generalizes across persuasion formats, with residual copy-mediated following as the honest boundary of the single-head approach.

### 4.4 Cross-model replication

Replicating the pipeline on Llama-3.2-1B and Qwen2.5-1.5B (600 known facts each, split 144/456 and 184/416 dev/held-out by relation; coarse and fine sweeps use the dev split only, capped at 300/150 facts) confirms the core structural and interventional findings, though the specialist taxonomy (Stage 4) was run only for Gemma-2-2B (Table 6). First, late-layer concentration holds in all three architectures: the most influential coarse layers are 20–25 of

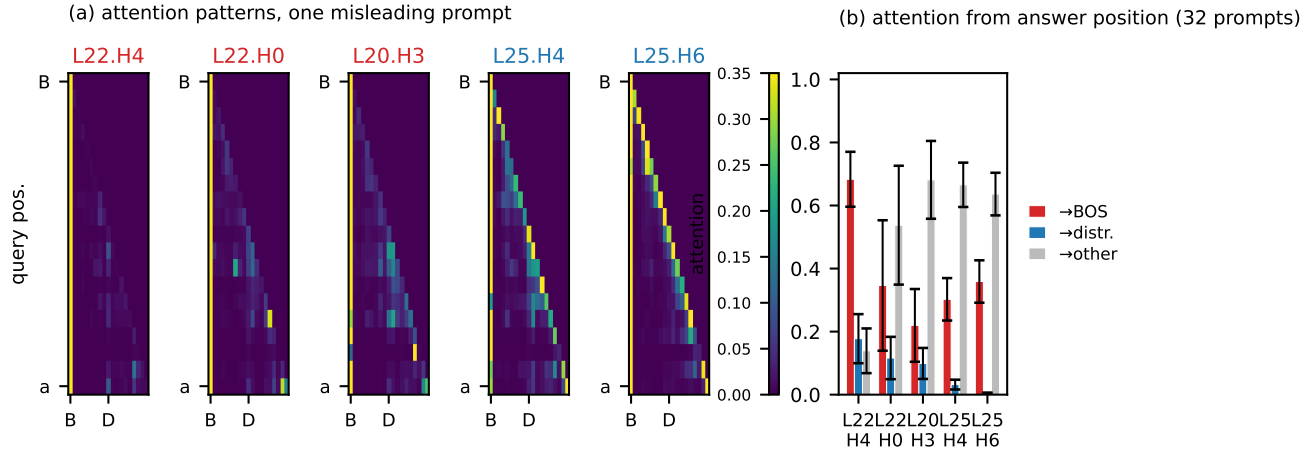



**Fig. 3** Attention of the five top heads. (a) Patterns on a representative misleading prompt (true: French, distractor: English); axis labels: B = ⟨bos⟩, D = distractor token, a = answer query. (b) Attention from the answer position to BOS / the distractor token / other positions, mean±SD over 32 misleading held-out prompts. L22.H4 (red) combines the strongest BOS concentration with direct distractor reading; dual-role heads (blue) read the distractor negligibly.

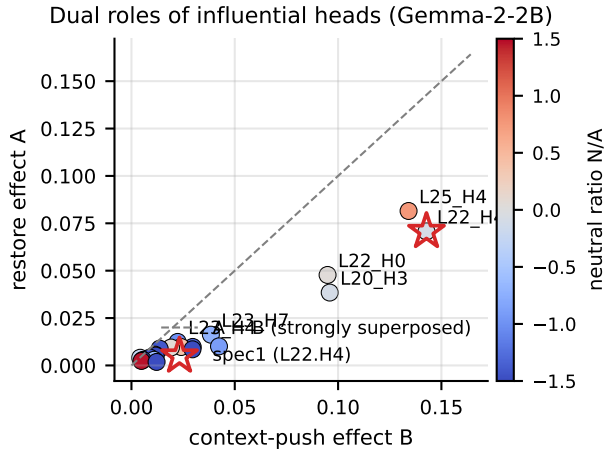



**Fig. 4** Dual roles of the top-20 heads: context-push effect $B$ vs. restore effect $A$, colored by neutral ratio $N/A$. Points near the diagonal are superposed; L22.H4 combines the strongest context conveyance with a minimal effect under the tested neutral control.

26 (Gemma-2-2B), 13–15 of 16 (Llama), and 22–27 of 28 (Qwen), and the normalized-depth profiles rank-correlate at 0.81 (Gemma-2-2B vs. each other model) and 1.00 (Llama vs. Qwen). Second, baseline distractor-following rates vary across models (6.6% and 14.9%, both lower than Gemma-2-2B's 19.3%; note that the splits are relation-disjoint but *not* item-identical—each model's known-fact filter retains different facts, 456/416 held-out facts for Llama/Qwen vs. 1,193 for Gemma-2-2B—so baseline differences mix persuadability with data composition). Third, and most importantly, the single-head knockout from each model's own dev ranking reduces distractor-following by 30% (Llama: 6.6%→4.6%, 95% CI 2.9–6.6, exact McNemar $p$=0.004) and 50% (Qwen: 14.9%→7.5%, CI 5.0–10.1, $p = 9 \times 10^{-10}$) with small accuracy changes (−1.3 and +0.8 points) and MMLU retention 1.01/1.00—a strong following reduction at near-zero accuracy cost, obtained without any per-model specialist analysis, suggesting the effect is a property of the patching-guided se-

**Table 4** Held-out interventions (Gemma-2-2B, misleading template T0; all values %). $n$=1,193 misleading + $n$=1,193 neutral items per config; follow rates carry fact-clustered bootstrap 95% CIs. Baselines differ from Table 1 because this table covers held-out facts only.

| config | follow↓ (95% CI) | truth↑ | neutral |
|---|---|---|---|
| none | 19.3 (17.0–21.5) | 74.4 | 93.5 |
| top-1 | 15.8 (13.8–17.9) | 66.8 | 82.6 |
| top-2 | 11.7 (9.8–13.5) | 69.9 | 80.1 |
| top-5 | 8.5 (7.0–10.1) | 44.7 | 50.9 |
| top-10 | 4.5 (3.4–5.8) | 32.0 | 39.6 |
| top-20 | 1.6 (0.9–2.3) | 19.4 | 25.9 |
| **spec1 (L22.H4 only)** | **12.6 (10.7–14.5)** | **79.7** | **92.0** |
| spec3 | 10.9 (9.1–12.7) | 63.3 | 68.1 |
| mean-abl. 2 | 11.2 (9.5–13.1) | 70.3 | 79.4 |
| mean-abl. top-5 | 8.8 (7.2–10.5) | 46.4 | 53.8 |
| mean-abl. spec3 | 11.4 (9.6–13.2) | 63.6 | 70.2 |
| topnorm-5 | 33.0 (30.4–35.8) | 59.1 | 85.9 |
| random-5 (5 seeds) | 12.6–22.2 | 44.8–78.8 | — |
| random-1 (5 seeds) | 5.2–12.2 | 81.9–88.3 | — |
| late-rand-1 (5 seeds) | 9.3–10.3 | 83.2–84.3 | — |

**Table 5** Capability under intervention (fixed 2,000-question MMLU subset; PPL over 1024-token WikiText-2 windows). A paired per-question rerun for L22.H4 gives 49.8%/49.4%, within bf16 run-to-run variation of the tabulated 50.0%/49.9%.

| config | MMLU | retention | PPL ratio |
|---|---|---|---|
| none | 50.0% | 1.000 | 1.00 |
| top-2 | 49.7% | 0.995 | 1.11 |
| top-5 | 50.5% | 1.011 | 1.20 |
| **spec1 (L22.H4 only)** | **49.9%** | **0.998** | **1.05** |
| spec3 | 50.4% | 1.009 | 1.16 |
| mean-abl. top-5 | 50.0% | 1.001 | 1.13 |

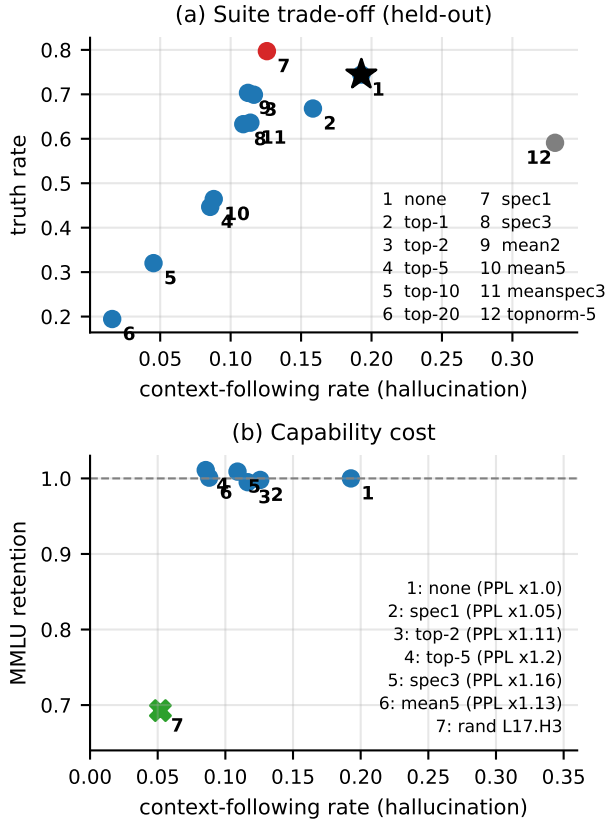



**Fig. 5** Trade-offs. (a) suite space: hallucination vs. truth; the specialist point (red P) attains the highest accuracy of all intervention configs while removing a third of distractor-following (top-2 removes more following, 11.7 vs. 12.6%, but at 9.8 points lower accuracy, so neither Pareto-dominates the other). (b) capability space: hallucination vs. MMLU retention (PPL ratios annotated).

**Table 6** Cross-model replication summary (misleading regime, held-out).

| | Gemma-2-2B | Llama-3.2-1B | Qwen2.5-1.5B |
|---|---|---|---|
| layers | 26 | 16 | 28 |
| top coarse layers | 20–25 | 13–15 | 22–27 |
| facts (dev/held-out) | 807/1,193 | 144/456 | 184/416 |
| coarse/fine sweep facts | 300/150 | 144/144 | 184/150 |
| knockout head (top-1) | L22.H4* | L15.H7 | L27.H2 |
| follow (none) | 19.3% | 6.6% | 14.9% |
| follow (top-1) | 12.6% | 4.6% | 7.5% |
| Δ truth | +5.3* | −1.3 | +0.8 |
| MMLU retention | .99–1.01 | 1.01 | 1.00 |

*Gemma-2-2B uses the Stage-4-selected specialist (L22.H4); the other models use their own dev-ranking top-1. All coarse/fine selection runs use the dev split only (capped at 300/150 facts; see Sect. 3). Gemma-2-2B retention spans the tested interventions (top-2/5, spec3, mean-abl.).

lection rather than of one architecture. Per-head restoration effects are smaller in the 1–1.5B models (0.04–0.06 vs. 0.08); this is consistent with a more distributed conflict circuit at smaller scale, though the data do not separate that reading from generally weaker per-head effects.

## 5. Discussion

**Mechanism.** The CM conflict is resolved late and attentionally: answer-position representations in the final layers carry most of what an intervention can restore, and the influential heads are conduits of the misleading context at least as much as carriers of memory ($B > A$ throughout). "Restoration" by patching therefore partly works by displacing a head's context conveyance rather than injecting memory—the neutral-control analysis makes this distinction measurable, addressing copy-vs-recall confounds.

**Attention-sink specialists.** The most surprising finding is the identity of the strongest conflict specialist: L22.H4, whose answer-position attention is dominated by the beginning-of-sequence token ($68 \pm 9\%$; the other specialists show only 22–35%). In the streaming-efficiency literature such sinks are treated as stabilizing but semantically inert no-ops [20]. Our results show that a sink head can be causally significant in the CM-conflict regime: L22.H4 conveys the strongest context push ($B$=0.143) and its removal improves accuracy at MMLU retention 0.998, PPL ratio 1.05. But $N \approx 0$ under one neutral-patching control only shows that this intervention had minimal effect in the tested condition, not that the head has no other function. The mechanism by which a BOS-attending head influences the conflict remains an open question—possible routes include value vectors at BOS, attention redistribution, or downstream effects—and although Fig. 3(b) aggregates 32 prompts, mechanism-level claims need larger samples. What the results do establish is that sink-focused heads should not be excluded from behavioral circuit searches.

**Design implication.** The dissociation between MMLU (no detected change on the tested subset) and perplexity (up 5–20%) suggests these heads support contextual integration that matters for language-modeling loss more than for benchmark accuracy; and the norm-selection failure warns that cheap importance proxies can worsen the targeted behavior.

**Limitations and future work.** (i) English-only, greedy completion-style evaluation; (ii) hallucination is operationalized as counterfactual override, not open-ended factuality; (iii) head selection is causal but the circuit description is coarse—path-level analysis is the natural next step; (iv) Gemma-2-2B is not the newest generation (chosen for tooling maturity and literature comparability); newer families and instruction-tuned variants are future work; (v) capability is proxied by MMLU/PPL subsets; (vi) the Stage-4 specialist taxonomy ran only for Gemma-2-2B, and entity-popularity effects remain open.

## 6. Conclusion

We mapped and exploited the CM-conflict circuit of small language models: late-layer attention heads with measurable superposition, among which specialist heads (identified via bidirectional patching and a neutral-control analysis in Gemma-2-2B) can be knocked out individually—distractor-

following falls by 30–50% relative on the standard templates (9% on the extreme adjacent-distractor template), accuracy rises for the Gemma-2-2B specialist (+5.3 points; replication models: −1.3 to +0.8), and MMLU retention remains near 1.00 on the tested subset—while norm-based selection backfires and the one capability-measured random head collapses MMLU (0.69). All selection was performed on dev relations with results reported on held-out relations, and the core findings (late-layer concentration, single-head intervention effectiveness) replicated across three architectures. Code, per-item records, and scripts are available from the author upon request.

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